

Discovering Patterns in U.S. Automobile Sales Using Transportation Statistics

M2: Initial Implementation Summary

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Dataset Overview

The dataset used for this project contains monthly transportation statistics including highway fatalities, rail incidents, freight transportation indices, and other transportation safety indicators. The goal of this analysis is to explore patterns in transportation activity and safety and identify meaningful structures within the data. The dataset contains several numeric indicators that describe transportation system performance over time.

Exploratory Data Analysis

Exploratory data analysis revealed several important characteristics of the dataset. First, many of the transportation safety metrics showed moderate correlations, particularly among rail-related fatality variables. Freight activity indicators such as the Transportation Services Index and the Truck Tonnage Index also exhibited positive relationships, suggesting that higher freight demand corresponds with increased trucking activity.

Distribution plots showed that several fatality-related variables are right-skewed, indicating that most months have relatively low fatality counts but occasional spikes occur. Boxplots also revealed potential outliers in highway fatality rates, which may correspond to unusually dangerous travel periods or seasonal effects.

These observations suggest that transportation safety and activity levels vary across time and may form distinct patterns that can be identified through clustering.

Preprocessing Decisions

Several preprocessing steps were required to prepare the data for mining. First, non-numeric columns were removed because clustering algorithms operate on numerical features. Columns containing only missing values were removed, as they do not contribute useful information to the analysis. Remaining missing values were imputed using the mean of each column to preserve the dataset size while maintaining reasonable estimates.

Because clustering algorithms rely on distance calculations, the numeric features were standardized using z-score normalization. This step ensures that features with larger numeric ranges do not dominate the clustering process.

Mining Technique and Results

Clustering was selected as the initial mining technique because the discovery question focuses on identifying natural groupings in transportation activity and safety patterns. The K-Means clustering algorithm was applied to the standardized dataset to discover natural groupings among the transportation indicators.

To determine the appropriate number of clusters, the elbow method was used. The elbow plot suggested that three clusters provided a good balance between simplicity and explanatory power. As a result, the final model used $k = 3$ clusters.

The clustering analysis revealed three distinct transportation patterns:

Cluster 0 – Moderate Transportation Activity. This cluster represents periods with moderate freight activity and moderate safety indicators. Transportation service indices and truck tonnage values fall near the middle range compared to the other clusters. Fatality metrics are also moderate, suggesting these observations represent typical transportation conditions.

Cluster 1 – Higher Safety Risk Periods. This cluster contains the highest average values for several fatality-related variables, including aviation and rail fatalities. However, freight and passenger transportation indices are lower compared to other clusters. This pattern suggests periods where transportation safety incidents were relatively elevated despite lower overall transportation activity.

Cluster 2 – High Transportation Activity Periods. The third cluster shows the highest transportation service indices and truck tonnage levels. These observations likely correspond to periods of strong freight and passenger transportation demand. Fatality metrics in this cluster are lower than in Cluster 1 but remain present, suggesting that increased activity does not necessarily correspond to the highest safety risks.

Overall, these results suggest that transportation data naturally separates into distinct operational regimes characterized by differences in activity levels and safety outcomes. Identifying these patterns helps reveal how transportation systems behave under different conditions and provides insight into periods where safety risks or freight demands are elevated.

Next Steps for M3

In the next milestone, additional mining techniques will be applied to deepen the analysis. Hierarchical clustering and DBSCAN may be used to explore whether alternative clustering methods reveal different patterns or structures within the data. Additionally, anomaly detection techniques may be used to identify unusual transportation events or extreme safety incidents that deviate from typical patterns.